

Troubleshooting Guide

Use this guide when something does not look right in the app or in the field. For safe battery handling, see [PowerSafetyNotes.pdf](#). For component behaviour, see [Hardware.pdf](#).

TG1. App issues

1. No GPS location in the app
 - a. Symptom: Map shows “No coordinates” or markers never appear.
 - b. Likely cause: The vest has no satellite fix, or the receiver has not yet written `lat`/`lon` to Firebase RTDB.
 - c. What people often do wrong: Expecting a fix indoors. Checking only the map while the receiver is off, not waiting through a cold start.
 - d. Fix / countermeasure:
 - i. Move the vest outdoors with open sky.
 - ii. Wait 1–3 minutes (cold starts can take several minutes).
 - iii. Confirm the receiver is on and Devices > last seen is updating.
2. Location is old / not updating
 - a. Symptom: Last seen updates but the pin barely moves, or coordinates look frozen.
 - b. Likely cause: Weak GPS, vest out of LoRa range, receiver Wi-Fi down, or Firebase not updating. The vest may also be sending last known coordinates after GPS lock is lost (see Vest section below).
 - c. What people often do wrong: Power-cycling repeatedly instead of fixing sky view or receiver connectivity, assuming online means GPS is current.
 - d. Fix / countermeasure:
 - i. Confirm last seen updates on the Devices screen.
 - ii. Move vest closer and more in line of sight to the receiver; move outdoors for GPS.
 - iii. Power-cycle only if needed after checking receiver Serial 115200 (maintainer).
3. Coordinates frozen but wet status still changes
 - a. Symptom: `wet` toggles in the app but lat/lon stay the same for a long time.
 - b. Likely cause: Vest firmware keeps transmitting last good GPS fix (`lastGoodFix`) even when live GPS stage drops (indoors, under cover).
 - c. What people often do wrong: Thinking GPS failed entirely, rain/LoRa may still work while position is stale.
 - d. Fix / countermeasure: Move vest outdoors until `gsn` shows `locked` again; expect fresh coordinates after a new lock.
4. Map shows ONLINE but Devices list says Offline (or vice versa)
 - a. Symptom: Same device appears online on the map but offline on Devices (or the opposite).
 - b. Likely cause: Not a bug, Live Map uses a 5 minute last-seen window for marker online ripple. Devices uses a 10-minute window for the Online/Offline label.

- c. What people often do wrong: Reporting inconsistent UI as a firmware fault.
 - d. Fix / countermeasure: Compare last seen timestamp, if within 5–10 minutes of now, behaviour is normal. If last seen is older, treat as connectivity/GPS/LoRa issue.
- 5. No data in the app / “No devices found under /vest”
 - a. Symptom: Empty Devices list or map overlay “No devices found”.
 - b. Likely cause: Empty RTDB path, security rules deny reads, or nothing uploaded yet.
 - c. What people often do wrong: Skipping Save settings after editing URL, signing in with wrong UID, receiver never flashed or powered.
 - d. Fix / countermeasure:
 - i. Verify Profile > Firebase RTDB URL (must start with `https://`) and tap Save settings.
 - ii. Sign in with a user whose UID matches Allowed UID and RTDB rules.
 - iii. Ensure receiver is running and logged into Firebase (USB Serial 115200).
- 6. Firebase init failed on the phone
 - a. Symptom: Full-screen “Firebase init failed” with Retry.
 - b. Likely cause: Bad URL, no network, or Firebase project misconfiguration.
 - c. What people often do wrong: Typo in database URL, using HTTP instead of HTTPS, no mobile data/Wi-Fi on phone.
 - d. Fix / countermeasure:
 - i. Re-check RTDB URL (`https://...firebase.database.app` style).
 - ii. Fix URL in Profile > Save settings > tap Retry on error screen.
- 7. “UID does not match” right after sign-in
 - a. Symptom: App signs you out immediately with expected vs actual UID.
 - b. Likely cause: UID typed on sign-in form is not the real Firebase Auth UID for that email.
 - c. What people often do wrong: Copying email, display name, or a custom ID instead of Authentication > Users > User UID in Firebase console.
 - d. Fix / countermeasure:
 - i. Copy UID from Firebase console or ask admin.
 - ii. Update Profile > Allowed UID to the same value > Save settings > sign in again.
- 8. Saved settings but app still empty
 - a. Symptom: URL and UID look correct in fields but map/devices stay empty.
 - b. Likely cause: Save settings not tapped after edits or URL/UID still wrong.
 - c. What people often do wrong: Assuming fields auto-save when leaving Profile.
 - d. Fix / countermeasure: Tap Save settings explicitly, confirm snackbar “Saved”, sign out and sign in if UID changed.
- 9. App won’t install (APK blocked)
 - a. Symptom: Android blocks installation.
 - b. Likely cause: Unknown sources not allowed for the app used to open the APK.

- c. Fix / countermeasure: Enable Allow from this source for the file manager/browser, then install again.
- 10. App opens but you can't sign in
 - a. Symptom: Sign-in errors or endless loading.
 - b. Likely cause: Wrong email/password, disabled user, or network error.
 - c. Fix / countermeasure: Read on-screen Firebase error text, reset password via admin if applicable.
- 11. Map is blank / grey tiles only
 - a. Symptom: Grey grid, no map imagery.
 - b. Likely cause: No internet on phone (OpenStreetMap tiles), or zero devices with coordinates.
 - c. Fix / countermeasure: Enable Wi-Fi or mobile data; if message says No coordinates, fix GPS (see above).
- 12. Wet alerts never fire
 - a. Symptom: Vest is wet in app but no notification.
 - b. Likely cause: Notifications disabled in Profile, Android blocked notifications, background worker cannot read RTDB, or no false to true transition occurred.
 - c. What people often do wrong: Expecting alert on every wet reading, disabling Android notification permission. Never opening app after install (background auth).
 - d. Fix / countermeasure:
 - i. Profile > Wet alerts on > Allow Android notifications > Send test notification.
 - ii. Open app signed in at least once so background tasks inherit session.
 - iii. Remember: alerts fire on dry to wet edge only, not on first app open baseline.
- 13. Wet alert on first app open (or no alert when already wet)
 - a. Symptom: No notification when opening app and device already shows wet.
 - b. Likely cause: Intentional, first observation sets baseline, only false to true transitions notify.
 - c. Fix / countermeasure: Dry the sensor until `wet=false`, then wet again to test; or use Send test notification in Profile.
- 14. Background wet alerts never fire (app closed)
 - a. Symptom: Foreground alerts work, none when app is swiped away.
 - b. Likely cause: WorkManager periodic task needs prior sign-in. Android throttles background work; poll interval too long.
 - c. What people often do wrong: Expecting instant background alerts, setting poll interval to maximum without reading clamp note.
 - d. Fix / countermeasure:
 - i. Sign in once; keep Wet alerts enabled.
 - ii. Profile > Background poll interval (clamped 15–240 minutes), lower for more frequent checks (Android may still throttle).
 - iii. Ensure RTDB rules allow reads for background session (email auth or anonymous fallback per deployment).

15. Remember me on a shared phone
 - a. Symptom: Next user sees previous credentials or wrong account.
 - b. Likely cause: Remember me stores email/password/UID locally (not encrypted vault).
 - c. Fix / countermeasure: Profile > Clear saved before handing off device, avoid Remember me on shared phones.

TG2. Vest unit issues

1. Signal lost / stale last seen (LoRa side)
 - a. Symptom: Last seen stops updating; Devices shows Offline.
 - b. Likely cause: Vest out of LoRa range, heavy obstruction, or receiver off.
 - c. What people often do wrong: Leaving receiver powered off, expecting range through thick concrete/metal.
 - d. Fix / countermeasure: Move vest closer, elevate receiver antenna, confirm receiver power LED/USB connected.
2. No GPS location `gsn: waiting\_for\_signal`
 - a. Symptom: RTDB or Serial shows stage 1 / waiting_for_signal, no lat/lon.
 - b. Likely cause: No NMEA bytes from GPS within 2 seconds (no sky, loose GPS power, failed module).
 - c. Fix / countermeasure: Outdoors with clear sky, wait cold start, maintainer checks GPS 3.3 V rail and UART wiring (see Appendix.pdf).
3. No GPS lock `gsn: no\_satellite\_lock`
 - a. Symptom: GPS bytes present but no fix, or fix too old.
 - b. Likely cause: Under cover, urban canyon, or fix age ≥ 2 s in firmware logic.
 - c. Fix / countermeasure: Open sky. hold still 1–3+ minutes, avoid metal roof overhead.
4. Device not powering on (vest)
 - a. Symptom: No activity after switches on.
 - b. Likely cause: Discharged battery, loose pack, or failed switch.
 - c. What people often do wrong: Assuming charge from yesterday is enough after long idle.
 - d. Fix / countermeasure: Recharge pack; inspect holder and switch (see PowerSafetyNotes.pdf).
5. LoRa INIT FAILED (vest Serial Monitor)
 - a. Symptom: Vest hangs at boot with “LoRa INIT FAILED” repeating 115200.
 - b. Likely cause: LoRa not powered on 3.3 V rail, wiring fault, or bad module.
 - c. Fix / countermeasure: Power cycle once, if persistent, escalate to maintainer (SPI/power: do not field-rewire unless trained).
6. Wrong device appears on map / mismatched ID
 - a. Symptom: Vest hardware label does not match `device\_01` vs `device\_02` in app.

- b. Likely cause: LoRa JSON `id` field maps to RTDB path (`id=1` to `device\_01`, zero-padded if id < 10).
- c. What people often do wrong: Flashing transmitter with wrong `VEST\_ID` constant.
- d. Fix / countermeasure: Confirm device ID with maintainer; select correct card in Devices / map filter.

TG3. Receiver / cloud issues

1. Receiver not uploading (Wi-Fi / Firebase)
 - a. Symptom: Serial shows LoRa RX but RTDB unchanged, app stale.
 - b. Likely cause: Wrong SSID/password, API key, database URL, or user credentials in `.ino`. SSL timeouts on poor links.
 - c. What people often do wrong: Editing secrets without re-flash; unplugging during backoff recovery.
 - d. Fix / countermeasure:
 - i. Serial Monitor 115200, read Wi-Fi / Firebase / RTDB lines.
 - ii. V2: wait through queue flush and backoff after transient outages.
 - iii. After editing secrets, re-flash ESP32.
2. LoRa OK in Serial but no RTDB writes (V2 specifics)
 - a. Symptom: “Received” logs but Firebase quiet.
 - b. Likely cause: 7 s quiet period after `Firebase.begin()`, token-not-ready waits 5 s, Wi-Fi/Firebase backoff active.
 - c. Fix / countermeasure: Wait 30–60 s after receiver boot; do not rapid power-cycle, check Wi-Fi RSSI in Serial if failures repeat.
3. Data gap after long Wi-Fi outage
 - a. Symptom: Hours of missing history after internet returns.
 - b. Likely cause: V2 queue holds 30 messages then drops oldest extended outage loses intermediate history (latest state still updates when flush resumes).
 - c. Fix / countermeasure: Fix Wi-Fi stability. accept that RTDB stores latest snapshot not full history.
4. `time\_my` missing in Firebase
 - a. Symptom: Last seen in app looks odd or falls back to other fields.
 - b. Likely cause: Receiver NTP never synced (`pool.ntp.org` blocked or no internet at boot).
 - c. Fix / countermeasure: Ensure receiver has internet at startup, reboot receiver once Wi-Fi stable, maintainer checks firewall/DNS.
5. Multiple vests, one always stale
 - a. Symptom: One device updates, another frozen.
 - b. Likely cause: Stale vest out of range, off, or wrong ID, shared queue still accepts packets but that vest not transmitting.
 - c. Fix / countermeasure: Check each vest power and proximity independently, verify each vest’s programmed `id`.

TG4. Rain sensor issues

1. Rain alert keeps showing (won't clear)
 - a. Symptom: `wet=true` persists in app.
 - b. Likely cause: Sensor plate still wet or high humidity on plate.
 - c. Fix / countermeasure: Dry plate and surrounding area, wait a few minutes for debounced dry state.
2. Rain alert seems false (no rain)
 - a. Symptom: Wet without rainfall.
 - b. Likely cause: Condensation, dirt, damp fabric touching plate, splash from ground.
 - c. What people often do wrong: Mounting sensor where sweat/rain runoff pools, ignoring condensation after cold storage.
 - d. Fix / countermeasure: Wipe plate, keep off wet surfaces (see Hardware.pdf).

TG5. Power issues

1. Repeated reboot loops (vest)
 - a. Symptom: Vest resets continuously.
 - b. Likely cause: Low battery, loose holder contact, or regulator undervoltage under LoRa TX load.
 - c. Fix / countermeasure: Power off, recharge fully, inspect pack (see PowerSafetyNotes.pdf).
2. Vest runtime much shorter than expected
 - a. Symptom: Dead after a few hours.
 - b. Likely cause: Cold weather, old cells, or left powered on overnight in bag.
 - c. Fix / countermeasure: Charge before use, switch off when idle, replace aged cell set (maintainer).

TG6. Debug logs

1. Serial Monitor on ESP32
2. Connect ESP32 (receiver or vest) to PC via USB.
3. Open Arduino IDE (or serial terminal).
4. Select correct COM port and board (ESP32 Dev Module family).
5. Open Serial Monitor at 115200 baud.
6. Press EN / RESET and capture log for support.

TG7. App-side logs

1. Profile > Logs, on-device ring buffer (not vest history).
2. Optional Upload logs to Firebase if deployment supports `appLogs/` path.

TG8. When to escalate to a maintainer

Provide:

1. Device ID (e.g. `device\_01`) and whether issue is vest, receiver, or app.
2. Symptom and approximate time (Malaysia time if known).
3. Serial capture 115200 (30–60 s after reset) for ESP32 issues.
4. Screenshot of Profile settings (redact password) and Devices last seen.
5. Whether receiver was powered and vest was outdoors for GPS tests.

Maintainers can refer for wiring and flash details in [Appendix.pdf](#) and [FirmwareSoftware.pdf](#).